

Project Design Phase

Problem–SolutionFit

Date	07 march 2026
TeamID	SWTID-2026-7607
ProjectName	MicroGoal
MaximumMarks	2Marks

Problem–SolutionFitOverview:-

TheProblem– SolutionFitensures that the identified problem matches real user needs and that the proposed backend solution effectively solves those issues before scaling further.

Purpose

- Addressthelackofa**secureandstructuredbackendsystem**formanagingpersonalgoals.
- Provideuserswitha**reliableAPI**to create,update,delete,andviewgoals.
- Ensure**secureauthenticationandauthorization**usingJWT.
- Improvedataprotectionbyrestrictingaccesstoauthorizeduseronly.
- Deliverascalablebackendarchitecturethatsupportsfuturefeatureexpansion.

ProblemStatement

Manyuserswanttomanagetheirdailyorlong-termgoalsdigitally,but:

- Existing systems are either not secure or overly complex.
- Some platforms do not protect user data properly.
- Unauthorized access to personal data is a major concern.
- Developers building simple goal management apps often struggle with authentication and error handling implementation.

As a result, users feel insecure about storing personal goals online, and developers face challenges implementing secure backend systems.

Solution

A **Node.js** and **Express**-based **Goal Management Backend API** that:

- Provides secure **User Registration and Login** functionality.
- Uses **JWT authentication** to protect API routes.
- Implements full **CRUD operations** for goal management.
- Ensures centralized **error handling middleware** for consistent responses.
- Stores data securely in **MongoDB** using structured models.
- Follows a modular architecture (Routes → Controllers → Models → Middleware) for scalability.